

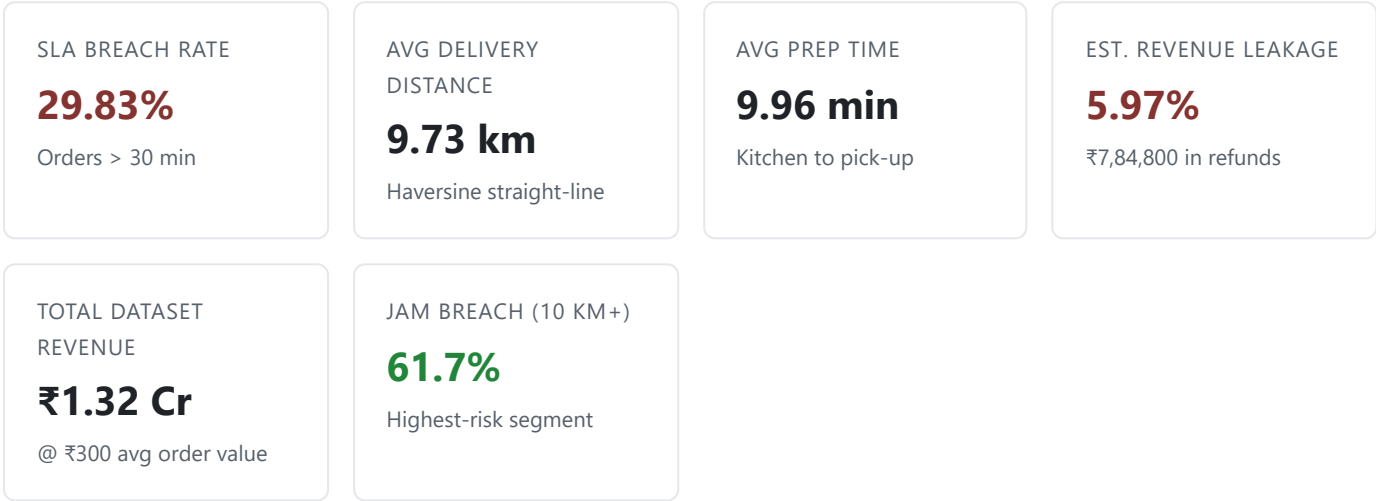
Zomato Delivery Analytics

End-to-end Exploratory Data Analysis & SLA Performance Report

Notebook: zomato.ipynb **Dataset:** zomato_delivery.csv **Raw Rows:** 45,584 | **After Cleaning:** 43,853

Columns: 20 raw → 28 engineered

KEY METRICS AT A GLANCE

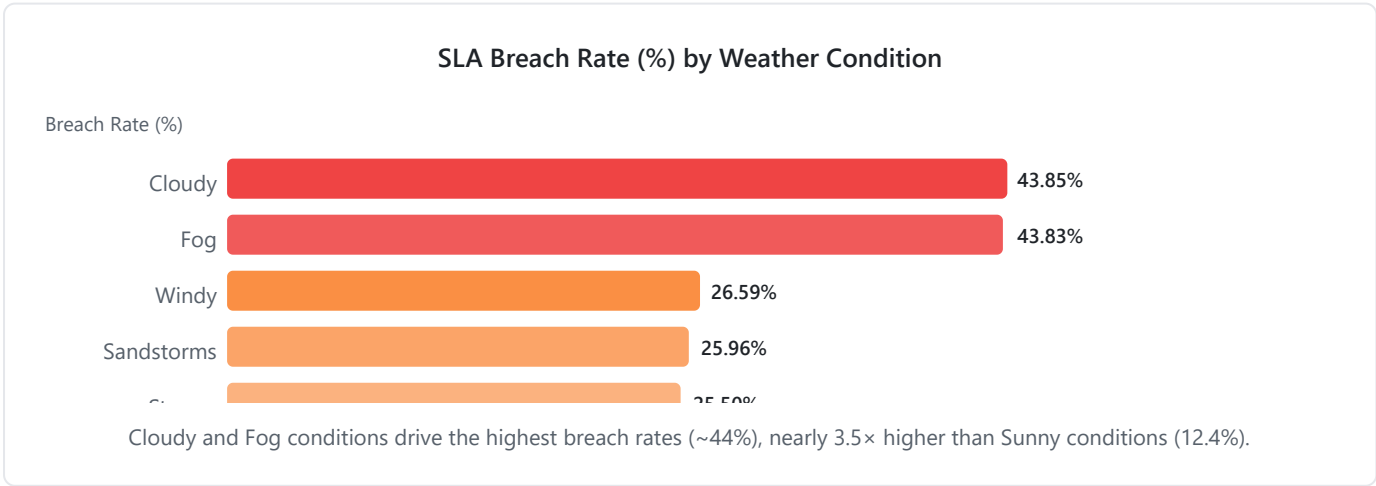


LIBRARIES USED



NumPy for vectorised math (Haversine, radians, arcsin). **Pandas** for all data wrangling, groupby aggregations, and pivot tables. **Matplotlib + Seaborn** for all chart and heatmap generation.

VISUALISATION 1 — SLA BREACH RATE BY WEATHER CONDITION



VISUALISATION 2 — SLA BREACH RATE BY TRAFFIC DENSITY

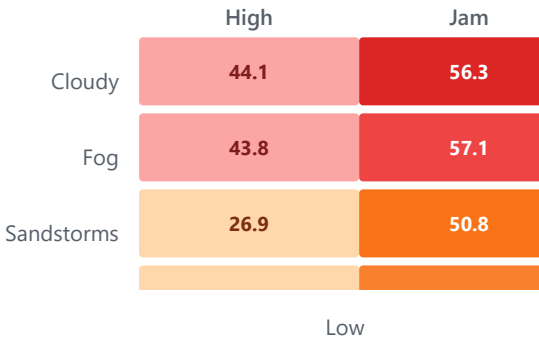
SLA Breach Rate (%) by Road Traffic Density



Jam traffic produces a 50.6% SLA breach rate — 6.4× higher than Low traffic (7.87%). High and Medium are nearly identical at ~33%.

VISUALISATION 3 — SLA BREACH HEATMAP: WEATHER × TRAFFIC DENSITY

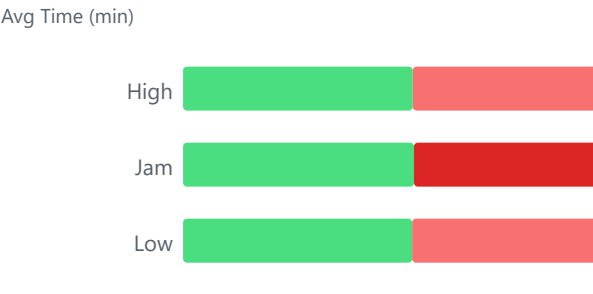
SLA Breach Rate (%) — Weather vs Traffic Density



Fog + Jam and Cloudy + Jam are the hottest cells (>56%). Sunny + Low traffic is the safest combination (3.6%).

VISUALISATION 4 — KITCHEN PREP VS ROAD TRANSIT TIME BY TRAFFIC

Fulfillment Time Breakdown: Kitchen Prep (green) vs Road Transit (red)

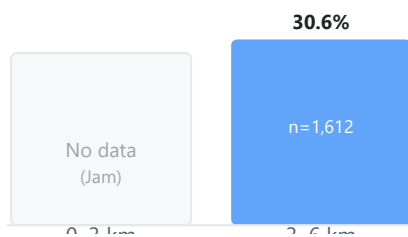


Kitchen prep time is nearly constant (~10 min) across all traffic tiers. Road transit is the variable — Jam adds ~10 extra minutes vs Low, pushing total time past the 30-min SLA.

VISUALISATION 5 — SLA BREACH RATE BY DELIVERY DISTANCE (JAM TRAFFIC)

SLA Breach Rate (%) by Distance Bucket — Jam Conditions

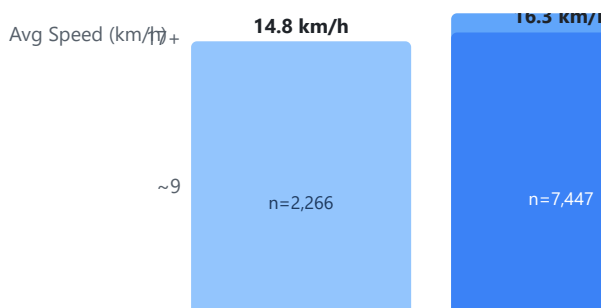
Breach Rate (%)



In Jam conditions, SLA breach nearly doubles from ~31% for short hauls (3–6 km) to 61.7% for long-distance orders (10 km+). The 10 km+ segment alone contains 7,850 orders.

VISUALISATION 6 — AVERAGE RIDER SPEED BY RATING GROUP

Average Delivery Speed Across Rider Rating Groups



High-rated riders (4.6–5.0) are ~17.5% faster than low-rated ones (<4.0). They also represent 67% of all filtered riders (n=21,837), making rating a strong proxy for operational efficiency.

VISUALISATION 7 — FINANCIAL IMPACT OF SLA BREACHES

Revenue Composition: On-Time Delivery vs SLA Breach Refund Cost



Nearly 6% of gross revenue is at risk from SLA breach refunds. Reducing the breach rate by just 10 percentage points would recover ~₹2.6 lakh in avoided credits.

DATA PIPELINE

1 Ingestion & Inspection

Load CSV (45,584 rows × 20 cols). Inspect dtypes. Strip whitespace from all object columns. Identify 8 columns with missing values.

2 Data Cleaning

Replace "NaN"/"null"/"" strings with np.nan. Parse Time_taken from mixed string format. Impute numeric cols (Age, Ratings, multiple_deliveries) with median. Fill categorical cols with sentinel values. Drop rows with missing GPS or Time_Orderd.

3 Geospatial Feature — Distance

Haversine distance computed between restaurant and delivery coordinates. Filter: 0 < distance < 50 km. Final shape: 43,853 rows.

4 Time Feature Engineering

Parse HH:MM timestamps. Derive prep_time_min (capped 0–45 min, overnight-wrap corrected). Derive transit_time_min and speed_kmh. Assign Time_of_Day slot.

5 SLA Labelling & Bucketing

SLA_Breached = 1 if Time_Taken_min > 30. Distance buckets: 0–3, 3–6, 6–10, 10+ km. Rider rating tiers: Low/Mid/High.

6 Analysis & Visualisation

Bar charts, stacked bars, heatmap, distance/breach chart, rider speed chart, financial donut chart across 7 analysis dimensions.

KEY FINDINGS

Overall SLA Breach Rate is 29.83% — nearly 1 in 3 orders misses the 30-minute target.

Jam + Fog/Cloudy is the worst combination — breach rates exceed 56–57%, found in the heatmap hot cells.

Distance is a multiplier in Jam conditions — breach rate jumps from 30.6% (3–6 km) to 61.7% (10 km+) in Jam traffic alone.

Road transit dominates delay, not kitchen prep — prep time is ~10 min across all traffic tiers; Jam adds ~10 extra transit minutes, crossing the SLA threshold.

Higher-rated riders are measurably faster — 4.6–5.0 rated riders average 17.4 km/h vs 14.8 km/h for <4.0 rated riders (+17.6%).

₹7,84,800 estimated refund exposure — representing 5.97% of total modelled revenue across the dataset.

RECOMMENDATIONS

Area	Recommendation
Traffic Routing	Dynamically route or throttle orders >6 km during Jam conditions. Real-time traffic feeds should gate order acceptance.
Rider Allocation	Assign 4.6+ rated (faster) riders to long-distance or adverse-weather orders to reduce breach probability.

Area	Recommendation
Order Batching	Reduce multiple_deliveries for Jam + long-distance segments; compounding transit time is severe.
Customer ETA	Dynamically inflate displayed ETA during Fog/Cloudy + Jam conditions to manage expectations and reduce perceived SLA failures.
Kitchen Monitoring	Prep time is stable at ~10 min but should be monitored during Evening Peak when demand spikes to avoid compounding road delays.

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